

Shruti Borkar

+1 (413) 510-7393 | shrutiborkar13@gmail.com | LinkedIn | Github | Portfolio

Data Engineer with 2+ years building production **ETL pipelines** and AI workflows on **Databricks, Spark, Airflow, and Delta Lake** – with hands-on **LangChain, RAG**, and **Agentic AI** expertise. Delivered 65% reporting reduction, 99.9% forecast accuracy, and 75% ML uplift in enterprise environments.

PROFESSIONAL EXPERIENCE

Five Edc Architects

January 2021 – May 2023

Data Engineer

- Reduced manual reporting by 65% by designing scalable SQL/Databricks ETL workflows and structured data models, improving data warehousing reliability and operational efficiency.
- Built an AI-powered knowledge graph on SCM data using graph-based ML and LLM-assisted entity resolution to extract vendor relationships from unstructured procurement documents – reducing RFP delays by 28%.
- Automated Power BI KPI pipelines backed by SQL + Python-based anomaly detection on financial forecast data, improving accuracy to 99.9% and eliminating 4 hours of weekly manual reconciliation across finance and operations.

Tata Consumer Products

May 2023 – July 2023

Data Science and AI Intern

- Built ETL pipelines with SQL, Python, and Alteryx to process 1M+ sales records with integrated dashboard components.
- Performed data profiling, normalization, and encoding via rule-based validation and statistical techniques to identify and correct data anomalies, improving downstream model reliability.
- Engineered a collaborative filtering + regression ML model for product recommendations using Scikit-learn and MLflow for experiment tracking, boosting accuracy by 75%; applied statistical feature selection (correlation analysis, mutual information) to identify top predictive variables across 50+ features
- Deployed AI-driven ThoughtSpot dashboards on SQL pipelines with NLP search, reducing reporting time by 90%.

PROJECTS

Netflix Genre Intelligence and Agentic AI Pipeline | Databricks, OpenAI API, LangChain, Power BI, Python, SQL

- Built an end-to-end Databricks pipeline to transform a 7,000+ title Netflix dataset – normalized 140+ country/genre rows, engineered gap metrics, and scaled K-Means clustering and GPT-4o recommendation tasks via OpenAI API
- Developed a LangChain RAG pipeline with Streamlit for natural language Q&A over genre data; layered a multi-agent Agentic AI system that monitors Apache Airflow DAGs, auto-diagnoses failures via LLM root cause analysis, and triggers self-healing actions – cutting on-call interventions by 70% with a Streamlit Q&A chatbot UI.
- Delivered Power BI dashboard with DAX KPIs to help identify genre gaps and prioritize localized content investments.

Medical Supply Chain Analytics Platform | Python, AWS S3, dbt Core, PostgreSQL, Streamlit, Neon

- Built end-to-end pipeline processing 100K+ records (e-commerce, logistics, fraud) — ingested CSV to AWS S3 (Parquet), transformed via dbt Core across 6 staging/mart models, enforced 20 data quality tests with 100% pass rate.
- Designed analytics layer surfacing operational risk across 103K+ orders — flagged 3,483 critical shipment delays and identified 3.6x fraud-rate disparity between Mastercard and American Express via dbt-modeled fraud signals mart
- Orchestrated three-stage ETL workflow (S3 → Postgres → dbt) using Prefect on Windows — replacing Airflow to eliminate WSL/Docker overhead and reduce orchestration code from a multi-file DAG project to a single Python flow.

TECHNICAL SKILLS

Languages: Python(Pandas, PySpark, NumPy), SQL(MySQL, PostgreSQL, BigQuery), R, DAX, MS Excel(Pivot, Macros)

AI / ML / LLMs: OpenAI API, LangChain, Agentic AI, RAG Pipelines, MLflow, Scikit-learn, NLP, Prompt Engineering

Data Platform: Databricks, Apache Spark, Apache Airflow, ETL Pipelines, Delta Lake, Snowflake, Data Warehousing

Cloud & BI: AWS (S3, Glue), GCP (BigQuery), Azure, Power BI, Tableau, ThoughtSpot, Amazon QuickSight, MATLAB

PUBLICATIONS & CERTIFICATIONS

- IJCA 2024:** Borkar et al. “Efficacy check of Haralick and Symmetry features for Skin Lesions Classification”
- Databricks:** AI Agent Fundamentals; SQL Analytics and BI on Databricks

EDUCATION

University of Massachusetts Amherst

Amherst, MA

M.S. in Business Analytics — GPA: 4.0 / 4.0

Expected May 2026

NMIMS University

India

B.Tech in Computer Science and Business Systems — CGPA: 3.57 / 4.0

May 2024